

Technology Stack

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Used to create a responsive and user-friendly interface for entering weather parameters and displaying flood prediction results.

2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, processes user input, loads the trained machine learning model, and returns prediction results efficiently.
3	Database / Data Storage	No Database (Pickle Model File - floods.save)	The project does not require a database since it only loads the trained machine learning model for prediction and does not store user information.
4	Cloud / Hosting / Deployment	Render	Used to deploy the Flask application online, making it accessible through a public URL with simple and cost-effective hosting.
5	Version Control & CI/CD	Git, GitHub	Git is used for version control, while GitHub stores the project repository and tracks changes for collaboration and deployment.

6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Scikit-learn, Pandas, NumPy, Pickle, Jupyter Notebook, VS Code	Scikit-learn is used to train the Random Forest model. Pandas and NumPy are used for data preprocessing and analysis. Pickle saves the trained model. Jupyter Notebook is used for model development and VS Code is used to build the Flask application.
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